ABIDES Daily Investor Baselines and Parameter Sweep Results

Belief-Aware RL Project

April 29, 2026

1 Research Goal and Report Roadmap

The central question is whether an RL trader can make better decisions by estimating a latent belief over the market participants it is trading against. In a real market, the agent does not observe each participant's strategy directly. It only observes public market data such as prices, spreads, order book imbalance, volume, and recent returns. The goal is therefore not to identify every individual trader, but to infer enough about the hidden market composition to choose better actions.

Formally, we want to learn a belief-conditioned trading policy:

$$b_t = P(\text{market regime or participant mix} \mid \text{public history up to } t), \qquad a_t = \pi(o_t, b_t),$$

where o_t is the usual market observation and b_t is an online estimate of whether the current market is more noise-dominated, momentum-dominated, or mean-reversion-dominated. A standard RL agent only sees o_t. A belief-aware agent sees both o_t and b_t.

1.1 What Are We Trying To Do?

We are trying to test whether public market data contains enough information to infer useful hidden structure about the agents generating the market. If the market is dominated by momentum traders, then following short-horizon pressure may be profitable. If the market is dominated by value or mean-reversion traders, then fading short-term moves may be better. A fixed policy averages across these regimes. A belief-aware policy should adapt as its estimate of the current regime changes.

1.2 What Experiments Are We Running?

This report studies two preliminary experiments that set up the belief-aware RL question.

First, we run a fixed baseline benchmark in the ABIDES daily-investor environment. Six hand-coded strategies trade in the same AAPL-calibrated market: `HOLD`, `BUY_HOLD`, `RANDOM`, `MOMENTUM`, `MR`, and `MR_INV`. This tells us which simple signals the default market rewards.

Second, we run one-at-a-time parameter sweeps over the background market and environment settings. These sweeps change the number and strength of value agents, momentum agents, noise agents, market-maker settings, order size, and timestep duration. The purpose is to check whether changing hidden market composition changes which strategy performs best.

1.3 How Do These Results Help?

The baseline and sweep results are not yet the final belief-aware RL experiment. They are the diagnostic layer that tells us whether belief estimation is worth doing. If every market composition

rewarded the same strategy, then estimating the hidden participant mix would add little value. But if different compositions favor different behavior, then a belief module has a real signal to exploit.

The current results support that motivation. The default AAPL calibration is strongly momentum-friendly: the `MOMENTUM` heuristic outperforms the mean-reversion heuristics by a wide margin. The value-agent sweep then shows the direction we care about: increasing fundamental anchoring makes mean-reversion more competitive and weakens momentum. This is exactly the kind of regime dependence a belief-aware policy should learn to detect from public history.

1.4 What Are The Next Steps?

The next step is to replace hand-coded strategy selection with a learned belief-conditioned policy. Concretely, we should:

1. Define a small set of latent market regimes, such as noise-dominated, momentum-dominated, value-agent-dominated, and mixed.
2. Generate ABIDES training data from each regime while exposing only public market observations to the learning agent.
3. Train a belief module that maps recent public history to a regime belief vector.
4. Train and compare two RL traders: a baseline policy $\pi(o_t)$, and a belief-aware policy $\pi(o_t, b_t)$.
5. Evaluate both trading performance and belief quality using P&L, Sharpe ratio, drawdown, win rate, and belief accuracy against the known simulated regime.

The rest of this document follows that flow. The setup sections explain the ABIDES market, the trading interface, and the heuristic policies. The results sections show which policies work under the default calibration and how those rankings shift under parameter sweeps. The conclusion translates those findings into the next belief-aware RL implementation plan.

We will then expand this to a multi-agent setting, where the agent is trading against a pool of other agents, each with their own unique strategy.

2 Metrics and Sweep Parameters

Before looking at the results, it helps to separate two kinds of quantities in this report. The **metrics** are what we measure after an episode finishes. The **sweep parameters** are the knobs we change in the simulator to create different market regimes.

2.1 Performance Metrics

Metric	Meaning
Final P&L	The final mark-to-market value minus the starting cash account. Positive P&L means the strategy ended the episode with more value than it started with. All P&L values in this report are measured in cents.
Total reward	The sum of per-step rewards over one episode. In this environment, reward is based on changes in mark-to-market value, so it should tell a similar story to P&L.
Sharpe ratio	A simple risk-adjusted score: mean final P&L divided by the standard deviation of final P&L across episodes. A higher Sharpe means the strategy earns more per unit of variability.
Win rate	The fraction of episodes where final P&L is non-negative. This is useful because a strategy can have a high average P&L but still lose often if its returns are very skewed.
Maximum drawdown	The largest peak-to-trough loss during an episode, measured as a fraction of peak mark-to-market value. Lower drawdown means the strategy is safer during the day.
Transaction-cost proxy	The sum of roughly half the spread paid on non-hold actions. It is not an exact exchange fee; it is a simple measure of how much active trading pays to cross the bid-ask spread.
Episode length	How many environment steps the episode lasts. Short episodes would indicate early termination from the drawdown kill switch; full-length episodes mean the strategy survived the whole simulated day.

Table 1: Metrics used to compare baseline strategies and sweep results.

2.2 Sweep Parameters

The sweep changes one parameter at a time. Each row below explains what that parameter controls and why it matters for belief-aware trading.

Parameter	What it changes
`num_value_agents`	The number of value agents in the background market. More value agents should create stronger fundamental anchoring and make mean reversion more plausible.
`lambda_a`	The arrival rate of value-agent trading opportunities. Higher values mean value agents show up more often, so the market should react more quickly when price deviates from fundamental value.
`kappa`	The mean-reversion strength assumed by value agents in their belief model. Larger values make value agents believe deviations from fundamental value close more quickly.
`kappa_oracle`	The true mean-reversion strength of the underlying fundamental-value process. This changes the actual environment dynamics, not just what value agents believe.
`fund_vol`	The volatility of the fundamental-value process. Higher volatility creates larger price opportunities but also more risk.
`num_momentum_agents`	The number of trend-following agents. More momentum agents should increase trend persistence and make momentum-style trading more attractive.
`num_noise_agents`	The number of random order-flow agents. Noise agents add volume and price jitter; they can also affect book depth because market makers size orders based on recent volume.
`mm_wake_up_freq`	How often adaptive market makers refresh their quotes. Slower refresh can leave stale quotes in the book; faster refresh can remove stale-quote opportunities.
`mm_spread_alpha`	How quickly the market-maker spread estimate adapts to recent market conditions. This affects the spread cost paid by active strategies.
`mm_window_size`	The market-maker lookback window for estimating spread and liquidity. This changes how stable or reactive the quoted market is.
`order_fixed_size`	The number of shares the daily investor buys or sells per market order. Larger orders increase exposure but can also increase slippage and spread cost.
`timestep_duration`	How much simulated time passes between the daily investor's decisions. Shorter timesteps allow more frequent reactions; longer timesteps produce coarser observations and fewer trading opportunities.

Table 2: Parameters varied in the one-at-a-time sweep.

3 Experimental Setup

The experiment evaluates the `markets-daily_investor-v0` ABIDES-Gym environment under an AAPL-calibrated `rmsc04` background market. The daily investor chooses among three actions at each step: market buy, hold, or market sell. Performance is measured by final mark-to-market P&L relative to the starting cash account.

The baseline benchmark uses 50 episodes per strategy with a 60 second timestep, starting cash of 1,000,000 cents, and fixed order size of 10 shares. All P&L numbers below are in cents.

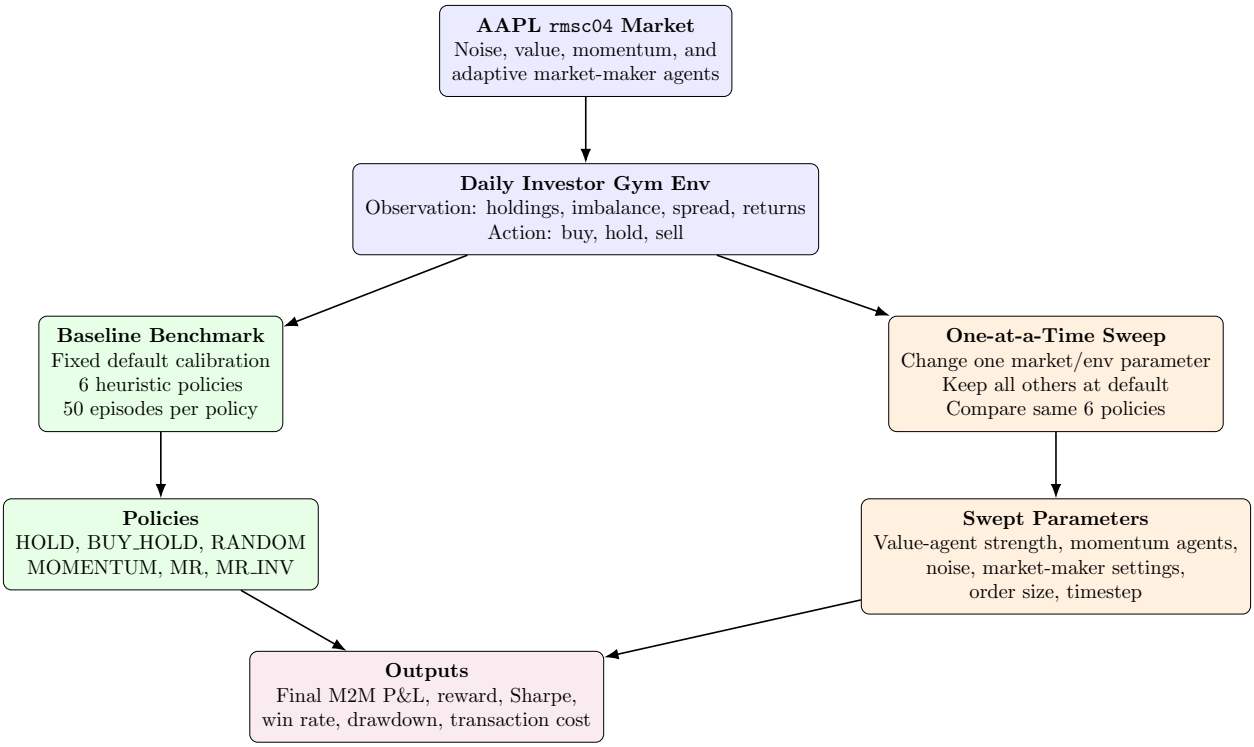


Figure 1: Experimental design. The baseline benchmark evaluates six fixed heuristics under the default AAPL calibration. The sweep keeps the same policy set but changes one environment or background-market parameter at a time to identify which market regimes favor momentum or mean reversion.

4 Setup Context

The daily investor is not trading against a simple price process. It trades inside an agent-based limit-order-book simulation. The background agents create order flow, liquidity, and price movement; the daily investor observes the resulting market state and submits one market order decision per environment step.

4.1 Background Market Agents

Agent type	Role in the simulated market
Noise agents	Submit random buy or sell orders. They create uninformed order flow and volume, which affects book depth, spread costs, and short-horizon price noise.
Value agents	Trade when the market price deviates from their belief about fundamental value. More active value agents tend to anchor prices and make the market more mean-reverting.
Momentum agents	Trade with short-horizon trends, typically buying after upward movement and selling after downward movement. More momentum activity can make price moves persist instead of immediately reverting.
Adaptive market makers	Continuously provide bid and ask liquidity around the current market. Their wake-up frequency, spread rule, and sizing determine how costly it is for the daily investor to cross the spread.

Table 3: Background agents in the `rmsc04` market configuration.

4.2 Daily Investor Interface

At each step, the daily investor receives a compact observation vector derived from the order book and its own portfolio, then chooses one of three actions.

Quantity	Location	Meaning
Holdings	`obs[0]`	Current signed share inventory. Positive means long, negative means short.
Imbalance	`obs[1]`	Bid volume divided by total visible bid and ask volume. Values above 0.5 indicate stronger bid-side pressure.
Spread	`obs[2]`	Best ask minus best bid, in cents. This is the immediate cost paid by market orders.
Direction feature	`obs[3]`	Difference between mid price and last transaction price, a short-horizon microstructure signal.
Recent returns	`obs[4:7]`	Lagged mid-price returns used by the momentum and mean-reversion heuristics.

Table 4: Observation features used by the daily investor environment.

The action space is deliberately simple: `0` submits a market buy, `1` holds, and `2` submits a market sell. Because these are market orders, active strategies pay spread and slippage whenever they trade. `HOLD` is therefore the zero-trading control: it should have exactly zero P&L in this setup.

4.3 Baseline Policies

The baseline policies are simple hand-coded trading rules. They are not trained RL agents; they are reference points that show which types of signals the market currently rewards.

Policy	Trading rule and interpretation
`HOLD`	Never trades. This is the sanity-check baseline: final P&L should remain zero because cash and holdings never change.
`BUY_HOLD`	Buys once at the start and then holds. This measures whether the simulated day has a directional long bias.
`RANDOM`	Chooses buy, hold, or sell uniformly at random. This is a noisy lower bound for active trading because it pays transaction costs without using a signal.
`MOMENTUM`	Follows order-book pressure. If bid-side imbalance is high, it buys; if ask-side pressure is high, it sells. The hypothesis is that current order-book pressure predicts near-future price movement in the same direction.
`MR`	Mean reversion. It fades recent mid-price returns: after a price drop it buys, and after a price rise it sells. The hypothesis is that short-horizon moves overshoot and then revert.
`MR_INV`	Mean reversion with an inventory cap. It uses the same signal as `MR` but stops adding to positions once inventory gets too large. This tests whether mean-reversion losses are caused mainly by uncontrolled position buildup.

Table 5: Hand-coded baseline policies evaluated in both the fixed benchmark and the parameter sweep.

4.4 How To Interpret the Sweep

The sweep changes one parameter at a time while keeping all other parameters at their default AAPL calibration. This isolates directional effects: for example, increasing `num_value_agents` tests whether stronger fundamental anchoring makes mean reversion more profitable, while increasing `num_momentum_agents` tests whether trend-following background flow makes momentum more profitable. Since the sweep plots show mean P&L with one standard deviation error bars, wide overlapping intervals should be interpreted as high uncertainty rather than a decisive ranking.

5 Baseline Strategy Results

Strategy	Mean P&L	Std. P&L	Sharpe	Win Rate	Mean DD	Txn. Cost
HOLD	0.00	0.00	0.000	100.0%	0.000	0.00
BUY_HOLD	-643.20	7336.97	-0.088	46.0%	0.009	1.25
RANDOM	6033.86	63751.45	0.095	58.0%	0.072	351.80
MOMENTUM	66171.66	131290.71	0.504	82.0%	0.122	371.30
MR	-57942.00	120705.01	-0.480	36.0%	0.140	462.57
MR_INV	-54810.98	26958.35	-2.033	2.0%	0.062	375.60

Table 6: Baseline performance over 50 episodes. Sharpe is computed as mean final P&L divided by standard deviation of final P&L across episodes.

The baseline result is clear: the default market regime is momentum-friendly and not mean-reversion-friendly. `MOMENTUM` has the highest mean P&L, the best Sharpe ratio, and an 82% win rate. The mean-reversion policies lose money on average. `MR_INV` reduces variance and drawdown relative to plain `MR`, but it is still consistently negative, with only a 2% win rate.

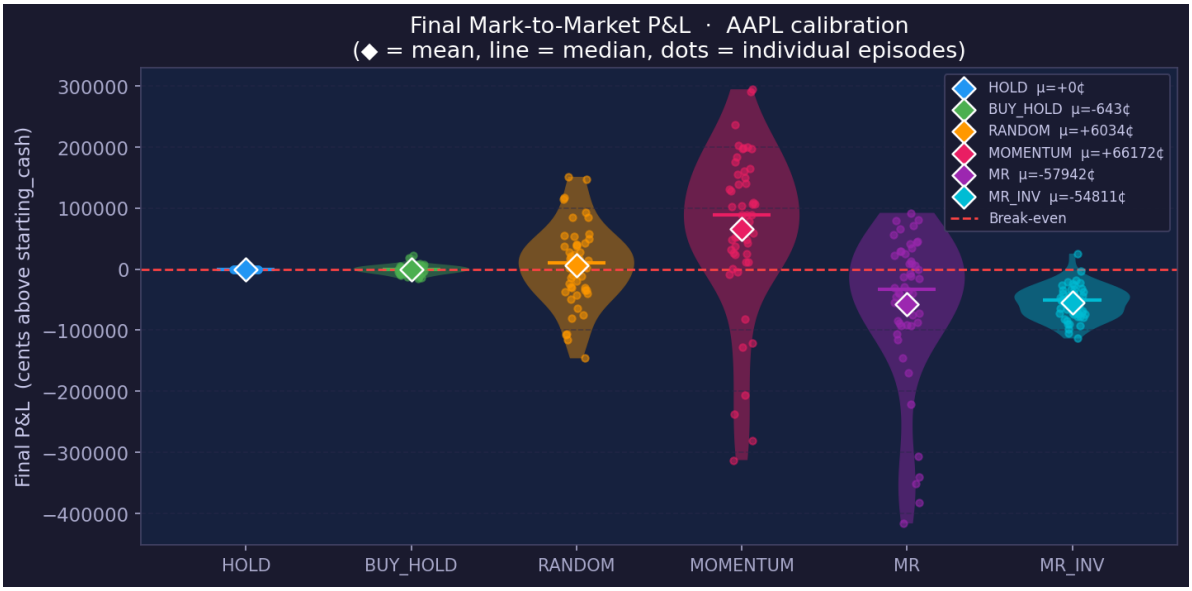
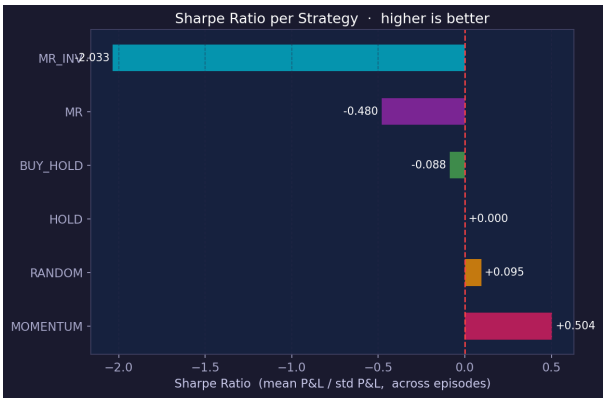
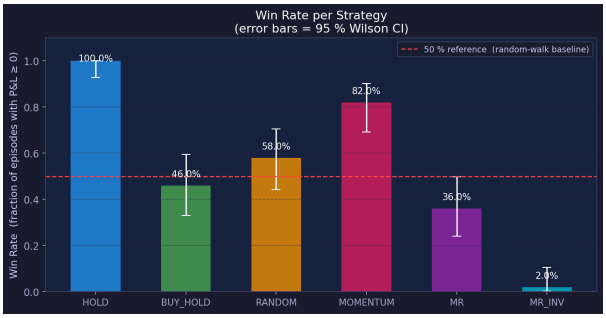


Figure 2: Final mark-to-market P&L distribution. Diamonds denote means, median lines show central tendency, and dots are individual episodes.

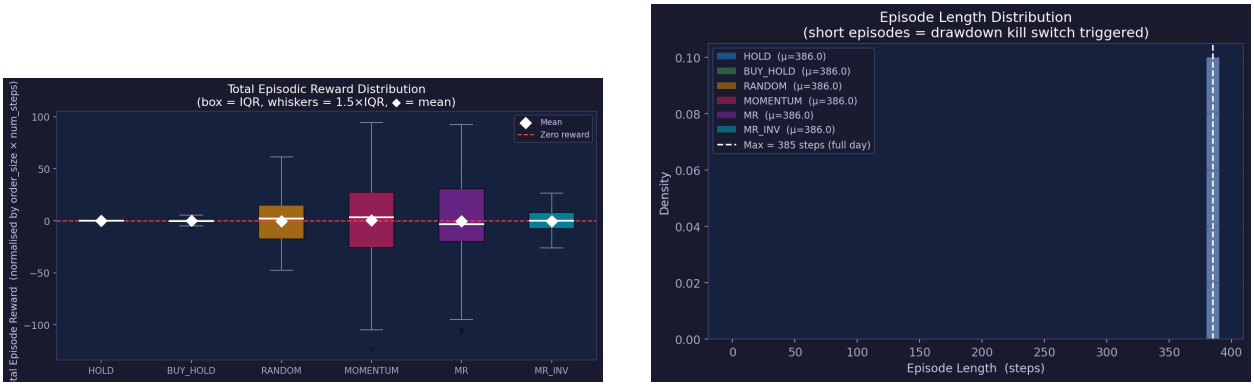


(a) Sharpe ratio



(b) Win rate

Figure 3: Risk-adjusted and directional success metrics.



(a) Episode reward

(b) Episode length

Figure 4: Reward and termination behavior. All strategies run for the full day, so the drawdown kill switch is not the main driver of the results.

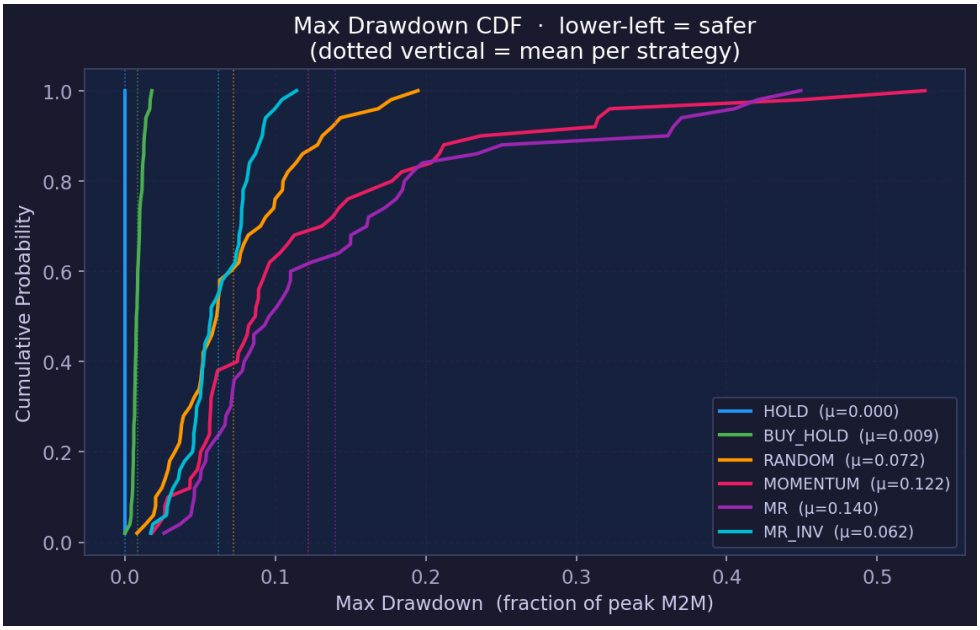


Figure 5: Maximum drawdown CDF. Curves farther left are safer. `MR_INV` has materially lower drawdown than `MR`, but its P&L remains negative.

5.1 Quick Guide to the Baseline Figures

Figure	What it shows	How to read it
Figure 2	The distribution of final P&L for each strategy across all episodes.	Look for strategies whose dots and mean diamond sit mostly above the red break-even line. `MOMENTUM` is the only strategy with a clearly positive mean and many winning episodes.
Figure 3	Two summary measures: Sharpe ratio and win rate.	Sharpe measures risk-adjusted performance; win rate measures how often P&L is non-negative. This figure is the fastest ranking of the policies: `MOMENTUM` is best, while `MR` and `MR_INV` are poor.
Figure 4	Reward distribution and episode length.	The episode-length plot confirms the strategies usually run for the full day, so the results are not mainly caused by early termination. The reward plot tracks the same story as P&L because reward is based on mark-to-market changes.
Figure 5	The distribution of worst within-episode losses.	Curves farther left are safer. `MR_INV` controls drawdown better than plain `MR`, but that risk control does not make it profitable.

Table 7: Reader's guide for the baseline figures.

6 Parameter Sweep Results

The sweep varies one parameter at a time and compares the same baseline strategies. The available plots show mean final P&L with one standard deviation error bars. The raw `sweep_results.json` file was not present alongside the plots, so these conclusions should be read as qualitative interpretations of the figures rather than exact statistical estimates.

Sweep	Main interpretation
`num_value_agents`	The cleanest qualitative result. As value-agent count increases, `MR` improves and `MOMENTUM` weakens, consistent with stronger fundamental anchoring creating a more mean-reverting market.
`num_momentum_agents`	The expected monotonic benefit to `MOMENTUM` is not clean in the plot; variance is large and most bars overlap zero.
`lambda_a`	Higher value-agent arrival rate appears favorable to `MR`, but the standard deviations are large.
`kappa`, `kappa_oracle`	Mean-reversion strength changes strategy rankings, but the plots are too noisy for a strong one-parameter claim.
`mm_wake_up_freq`	The 60s setting looks most favorable to `MR`; 120s is worse for `MOMENTUM`. Error bars remain wide.
`order_fixed_size`, `timestep_duration`	No stable monotonic transaction-cost story is visible from the current sweep.
`fund_vol`, `num_noise_agents`, market-maker parameters	These change opportunity and noise, but current sweep variance is large enough that results should be treated as directional only.

Table 8: Qualitative interpretation of the one-at-a-time parameter sweep plots.

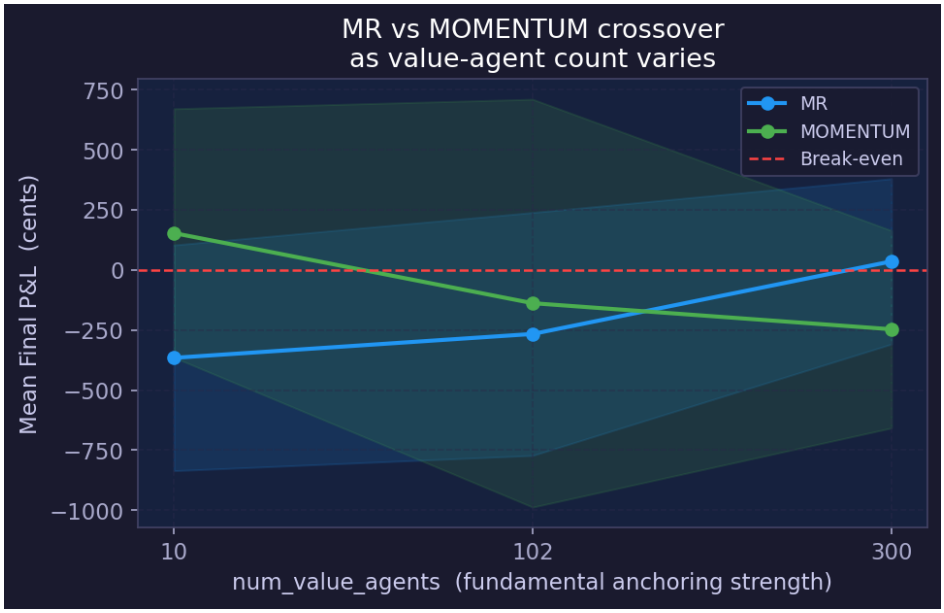
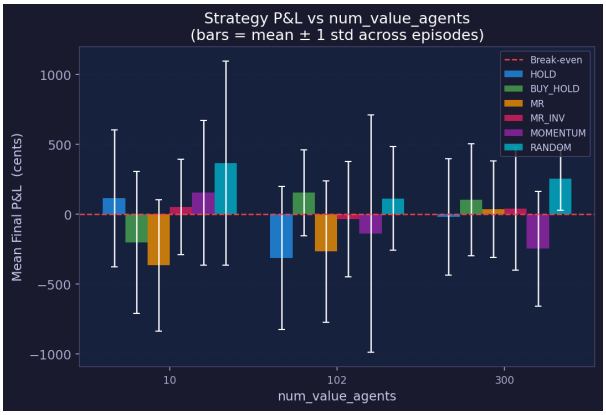
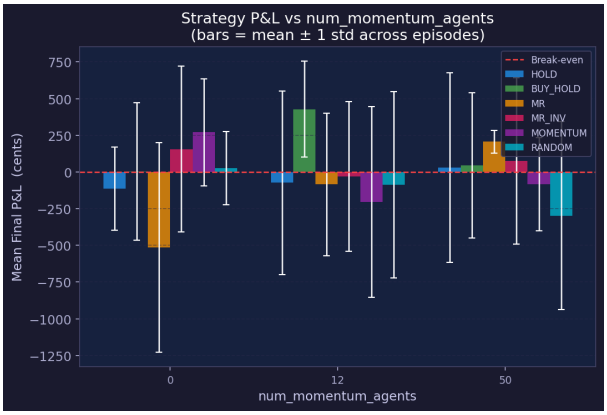


Figure 6: `MR` versus `MOMENTUM` as value-agent count varies. This is the most important sweep figure: more value agents move the market toward a mean-reverting regime.

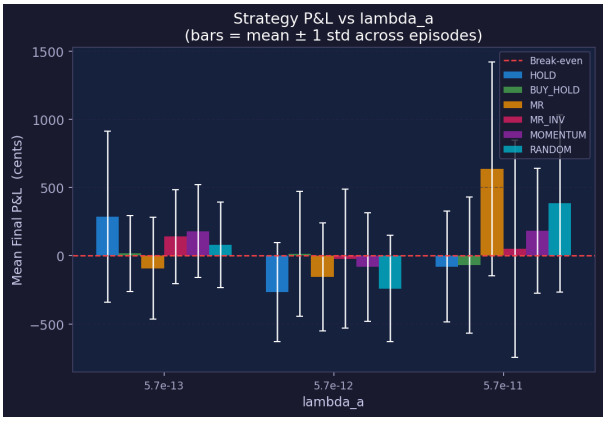


(a) Value agents

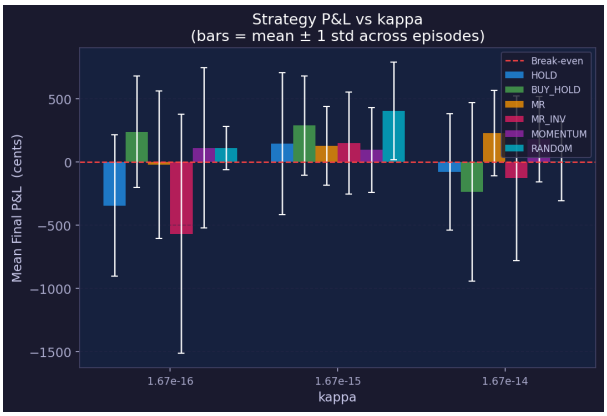


(b) Momentum agents

Figure 7: Agent-population sweeps.

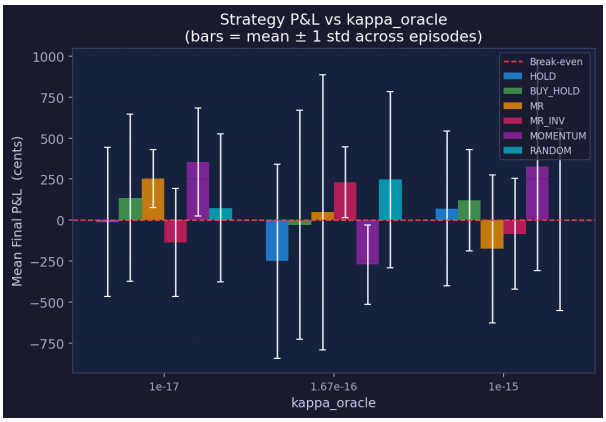


(a) `lambda_a`

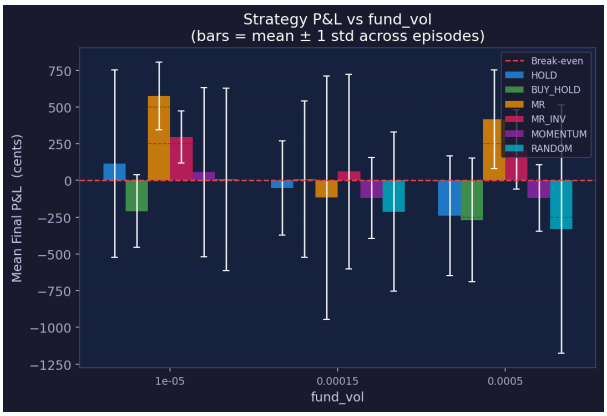


(b) `kappa`

Figure 8: Value-agent belief and arrival sweeps.

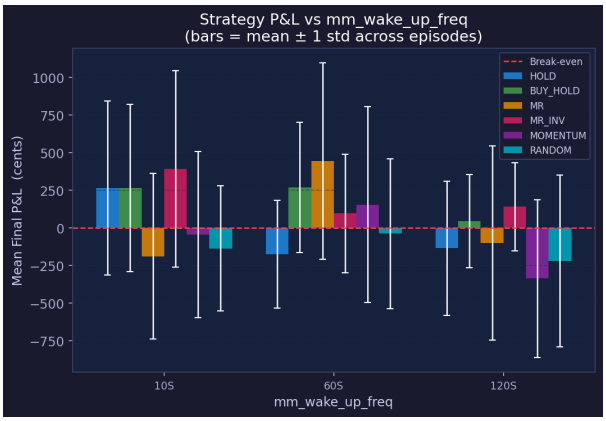


(a) `kappa_oracle`

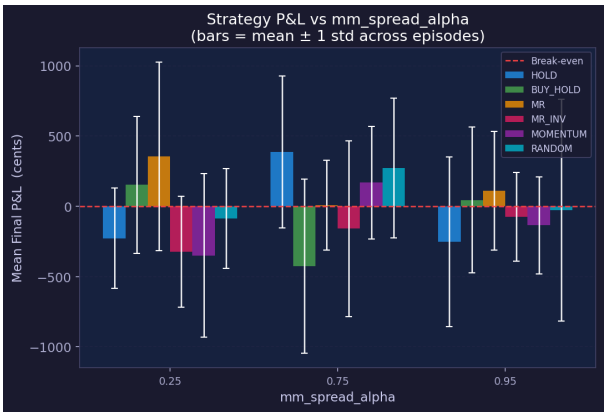


(b) `fund_vol`

Figure 9: Oracle fundamental-process sweeps.

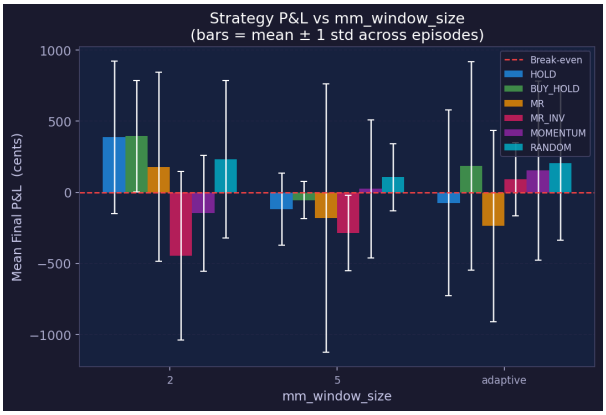


(a) Wake-up frequency

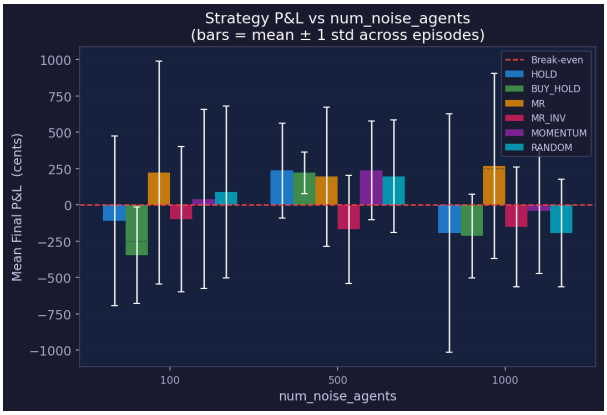


(b) Spread alpha

Figure 10: Market-maker timing and spread-adaptation sweeps.



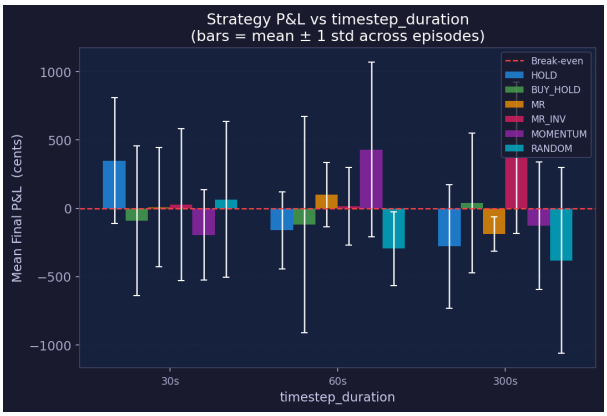
(a) Window size



(b) Noise agents

Figure 11: Liquidity and noise sweeps.

(a) Order size



(b) Timestep duration

Figure 12: Environment-level action-frequency and order-size sweeps.

6.1 Quick Guide to the Sweep Figures

Figure	What it varies	What to take away
Figure 6	Only the number of value agents, comparing `MR` against `MOMENTUM`.	This is the most important sweep figure. More value agents make the market more fundamentally anchored, which improves `MR` and weakens `MOMENTUM`.
Figure 7	The number of value agents and momentum agents.	These plots ask whether changing the hidden trader population changes which strategy works. The value-agent trend is the cleanest evidence that regime composition matters.
Figure 8	Value-agent arrival rate `lambda_a` and value-agent belief strength `kappa`.	These parameters control how aggressively value agents react to deviations from fundamental value. The plots are noisy, but they help identify candidate mean-reverting regimes for future RL training.
Figure 9	The true fundamental process: `kappa_oracle` and `fund_vol`.	These change how the underlying fundamental value moves. Larger changes can create more opportunity, but also more risk; wide error bars mean the current sweep is not decisive.
Figure 10	Market-maker wake-up frequency and spread adaptation.	These parameters control liquidity refresh and spread behavior. They matter because every active policy crosses the spread when it buys or sells.
Figure 11	Market-maker window size and the number of noise agents.	These plots probe liquidity and noisy order flow. They mostly show high variance, so they should be treated as diagnostic rather than conclusive.
Figure 12	Order size and timestep duration.	These are environment-level controls. They change how often the agent trades and how large each trade is, so they affect transaction costs and signal granularity.

Table 9: Reader's guide for the parameter sweep figures.

7 Takeaways

The results should be read as a chain of evidence:

1. **First, the default market has a clear winner.** In the baseline table, `MOMENTUM` has the highest mean P&L, the best Sharpe ratio, and the highest non-trivial win rate. The mean-reversion strategies, `MR` and `MR_INV`, lose money on average. This tells us that the default AAPL-calibrated market rewards short-horizon momentum more than mean reversion.

2. **Second, this is not just because all active trading works.** `RANDOM` trades actively but performs much worse than `MOMENTUM`. That means the momentum strategy is using a useful signal, not merely benefiting from taking more trades.

3. **Third, changing the market composition changes the story.** In the value-agent sweep, increasing `num_value_agents` makes `MR` improve and makes `MOMENTUM` weaken. This is important: it suggests that different hidden market regimes favor different trading behavior.

4. **That is exactly why belief estimation is useful.** If the best action depends on whether the market is momentum-heavy or value-agent-heavy, then an RL trader should not use one fixed policy for every regime. It should estimate the current regime from public market history and condition its policy on that belief.

5. **The immediate next step is to build those regimes explicitly.** We should train and evaluate two agents across several known ABIDES market compositions: a standard RL trader that sees only market observations, and a belief-aware trader that also receives an estimated regime vector. The belief-aware trader should be judged by both trading performance and how well its inferred beliefs match the known simulated market composition.

The main caveat is that the sweep plots are still noisy. They are strong enough to motivate the next experiment, especially the value-agent result, but most individual parameter effects need more episodes or the raw sweep table before they should be treated as final statistical claims.

This also translates to the main multi-agent experiment. The trading agent is not trying to identify one named opponent. Instead, it observes the public footprint of many hidden agents acting together: prices, spreads, imbalance, and returns. The belief module should therefore estimate the current multi-agent market composition, such as "mostly momentum," "mostly value," or "mixed." If different compositions favor different trading behavior, as the sweep suggests, then conditioning the RL policy on that belief is the right multi-agent analogue of estimating a partner type in a cooperative game.

8 Next Steps

The next iteration should make the sweep results more signaling. Right now, many sweep plots have large error bars that overlap zero and overlap each other. Those plots are useful diagnostics, but they are not strong enough to support precise claims about every parameter.

- **Most noisy sweeps.** The least conclusive sweeps are `num_momentum_agents`, `kappa`, `kappa_oracle`, `fund_vol`, `num_noise_agents`, the market-maker parameters, `order_fixed_size`, and `timestep_duration`. Their error bars are wide and the strategy rankings are not stable across parameter values.

- **Most useful sweep so far.** The clearest signal is `num_value_agents`. As value-agent count increases, `MR` improves while `MOMENTUM` weakens. This directly supports the idea that hidden market composition changes which trading behavior is best.

- **Improve statistical signal.** Run more episodes per parameter-value-strategy cell, save the raw `sweep_results.json`, and plot confidence intervals rather than standard deviation bars. Confidence intervals answer the question "how certain are we about the mean?" more directly than standard deviations.

- **Use stronger regime definitions.** Instead of only changing one parameter slightly at a time, create explicit regimes: a momentum-heavy regime, a value-agent-heavy regime, a noise-heavy regime, and a mixed regime. This should produce cleaner differences for the belief module to learn.
- **Simplify the diagnostic plots.** For the next sweep, focus on `MOMENTUM`, `MR`, `MR_INV`, and `HOLD`. The full six-strategy plots are crowded, while the belief-aware RL question is mainly about whether the market favors momentum or mean reversion.
- **Plot differences from default.** In addition to raw P&L, plot each strategy's change in mean P&L relative to the default parameter value. This will make it easier to see whether a parameter actually moves the market regime.

In short, I will keep iterating on this by turning the current broad sweep into a smaller, targeted regime-identification experiment. The goal is to produce clearer market regimes first, then train the belief-aware RL agent to infer those regimes from public observations.